



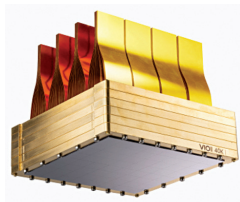
gateway, and battery junction box controller, allowing OEMs to monitor battery health in real time without added sensors or software. By embedding EIS in hardware with an on-chip discrete Fourier transform, the chipset enables impedance measurement during charging and load changes, helping OEMs detect ageing effects, temperature gradients, and microshort circuits, and reduce system complexity.

NXP Semiconductors

<https://www.nxp.com>

QPU architecture

The quantum industry has been stuck at QPUs with around 100 qubits, forcing companies to link many small processors at high cost to pursue larger-scale computing. QuantWare's VIO



40K addresses this bottleneck with its VIOTM scaling architecture, enabling QPUs with 10,000 qubits, 100

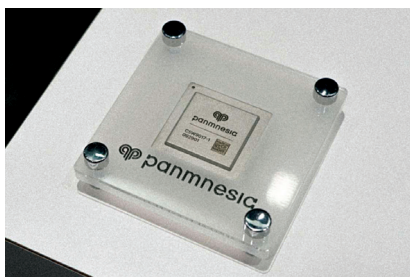
times larger than current devices, in a smaller package. VIO 40K uses 3D scaling with 40,000 input-output lines and high-fidelity chiplet connections, delivering more compute per dollar and per watt. It is available to any organisation working with superconducting qubits, allowing them to build much larger, more powerful quantum processors.

QuantWare

<https://quantware.com>

Fabric switch

Panmnnesia Inc. announced that samples of its PCIe 6.4/Compute Express Link (CXL) 3.2 Fabric Switch are now available. This is the first CXL switch chip that supports port-based routing (PBR). The switch supports both Peripheral Component Interconnect



Express (PCIe) Generation 6 and CXL 3.2 on a single chip and is compatible with earlier PCIe and CXL versions. It complies with all CXL 3.2 and PCIe 6.4 rules, supports both PBR and hierarchy-based routing, and integrates easily with other systems. The switch helps build flexible systems that reduce CAPEX and OPEX while delivering high performance for AI tasks such as deep learning recommendation models, LLMs, and retrieval-augmented generation.

Panmnnesia Inc.

<https://panmnnesia.com>

CMOS image sensor

Cars need cameras that are fast, reliable, and small. They also need to detect bright objects and work even when parked. Sony's IMX828 addresses these needs. It is the first automotive CMOS image sensor



with a built-in MIPI A-PHY interface, making it the first sensor to integrate this high-speed standard directly. The sensor has an 8-megapixel effective resolution and the industry's highest HDR level. It eliminates the need for external serialiser chips, reducing cost, board size, power consumption, and heat generation, while improving resistance to external noise-induced errors. Enhanced HDR improves recognition of high-luminance objects, such as red LEDs. A low-power parking surveillance function detects moving subjects. At the same time, when the vehicle is parked, it uses under 100mW by capturing low-resolution, low-frame-rate images and switching to normal imaging when motion is detected.

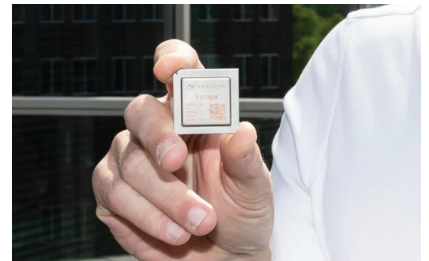
Sony

<https://www.sony.co.in/homepage>

AI processor unit

Organisations running generative AI and computer vision applications face

high costs and low efficiency with existing AI processors. Axelera AI addresses this with Europa, an AI processor unit that delivers performance at lower cost. Europa supports multi-



user workloads and compute-intensive, multimodal AI inference across edge devices and enterprise servers. It can deliver up to 629 TOPS at INT8 precision and a peak of 4915 GOPS. Its high-throughput design provides 3 to 5 times better efficiency per watt and per dollar compared to current solutions. Europa is available in multiple form factors, uses less energy, and is designed to make AI applications more accessible to organisations.

Axelera AI

<https://axelera.ai>

NFC chip

Setting up smart homes is often slow and complex. STMicroelectronics addresses this with the ST25DA-C, a secure NFC chip that enables users to add lights, cameras, access controls,



or other IoT devices to a Matter-based network in one step by tapping their phone. It is the first commercial chip supporting the latest Matter updates, improving security, reliability, and ease of use. Compared to Bluetooth or QR-code setups, it speeds up commissioning, allows multiple devices to be installed at once, and can perform cryptographic tasks using power harvested from the NFC field, enabling the setup of unpowered devices.

STMicroelectronics

<https://www.st.com>